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Excel

Basics

① Excel file = workbook

→ contain one or more sheets

② worksheet = sheet = All the cells

→ contain cells arranged in rows and columns.

③ Sheet tab = Name of worksheet

→ at the bottom, can edit worksheet names

④ Selection cursor 

→ click middle of a cell, hold & drag

⑤ Move cursor 

→ appears when mouse is on edge of selected cell or range

→ used to move content of cell(s) to another location

⑥ Fill Handle cursor 

→ appears at bottom right corner

→ used to copy or fill cell values / formulas by dragging

⑦ Resize Column / Row cursor ( or )

→ appears when mouse is b/w column letters / row numbers

→ used to resize column width or row height by dragging

⑧ Ribbon

→ menu bars containing tools & features

→ divided into tabs contains group of commands

⑨ Mini tool bar

→ floating menu appears when you right click on a cell or highlight text

→ quick formatting

Commands

① Ctrl + Enter

- keeps the cursor in same cell after typing
- used to enter same value in multiple selected cells

Ex:- 1. select many cells
2. type a value
3. Press ctrl + Enter

} same value

② Tab

- moves cursor to the next cell on right
- used for fast data entry in rows

Ex: A1 → Tab → B1 → Tab → C1

③ Enter

- moves cursor to next cell down
- used for fast data entry in columns

Ex: A1 → Enter → A2 → Enter → A3

④ Ctrl + S

- used to save Excel file

⑤ Ctrl + *

- selects all connected cells

Use: select full dataset quickly

⑥ Ctrl + 1

- opens format cells window

- can change number format, date format, font, border, alignment

⑦ F4 (fn + fu)

- repeats last action

- works for formatting, delete, insert, color... etc

- does not repeat typing text

Ex: Bold → F4 repeat Bold

Delete → F4 repeat Delete

→ Also used to lock cell reference in formula

ex: write =A1 in C2 cell

then press $F_n + F_4$

it will become = \$A\$1

⑤ $Ctrl + C \rightarrow$ copy

$Ctrl + F \rightarrow$ find data

$Ctrl + V \rightarrow$ Paste

$Ctrl + H \rightarrow$ find & replace data

$Ctrl + X \rightarrow$ cut

$Ctrl + Z \rightarrow$ Undo

$Ctrl + Y \rightarrow$ Redo

$Ctrl + A \rightarrow$ select All

⑥ Double click (Ribbon Hide) [$Ctrl + F_n + F_i$]

→ double click on any ribbon tab to hide the ribbon

→ when hidden, only tab names are visible

Use: more screen space

⑦ select row / column.

→ Shift + spacebar : select entire row

→ Shift + Ctrl + spacebar : select entire column.

⑧ Navigation

→ Alt + Enter : Start a new line within a cell.

→ Ctrl + Arrow keys : Move to the edge of the current data region.

→ Ctrl + Home : Move to the beginning of the worksheet

→ Ctrl + End : Move to the last cell with content

→ Ctrl + Page Up/Down : Move between worksheet tabs.

→ Ctrl + Shift + U : Increase height of formula bar

Default Alignment

Text → left

Number → right

Date → right

Boolean → centre

Data Entry Trick

- Select the cells where the data needs to be entered before starting
- This way, when you press Enter after in the last cell of a column, cursor doesn't move down to the cell below. Instead, it moves to the first selected cell of next column.

Formulas

- Used to calculate values & always starts with =

Ex:- = A1 + A2

= A1 * B1

= SUM(A1:A5)

① Basic Formulas

→ SUM : = SUM(A1:A5)

→ AVERAGE : = AVERAGE(A1:A5)

→ MIN : = MIN(A1:A5)

→ MAX : = MAX(A1:A5)

→ COUNT : = COUNT(A1:A5) [counts numbers]

→ COUNTA : = COUNTA(A1:A5) [counts non-empty cells]

② Logical Formulas

③ Text Formulas

④ Lookup Formulas

⑤ Math Formulas

⑥ Count Formula

later

Selecting full data quickly:

- ① Ctrl + Shift + Down Arrow (selects data till last row)
 - ② Ctrl + Shift + Right arrow (selects data till last column)
- } start with top-left

Copying data from one sheet to another using Ctrl+C & Ctrl+V (same workbook)

1. select data
2. press Ctrl+C
3. Go to another sheet
4. press Ctrl+V

Copying data from one excel sheet to another using formula (same workbook)

1. Go to new sheet
2. Type =
3. Go to old sheet
4. Select cell / range / full data
5. press enter

Copying data from one sheet to another using move or copy (same workbook)

1. Right click on sheet tab
2. select move / copy
3. To book :
 - same book
 - so don't change
4. select before which sheet the selected sheet should copy / move
5. click ok.

Ctrl+C ← Ctrl+V

Copying data from one sheet to another using formula (different workbook)

1. open both excels
2. select data in first workbook
3. press Ctrl+C
4. Go to second workbook
5. Press Ctrl+V

Copying data from one sheet to another using formula (different workbook)

1. Open both workbooks
2. Go to new work book

3. Type =

4. Go to other workbook

5. select cell / range / full data

6. press Enter.

Copying data using Move or Copy (different workbook)

1. open both workbooks

2. Right click sheet tab

3. click move or copy

4. In 'To book', choose another workbook

5. select before which sheet the selected sheet should move or copy.

6. click ok.

Ribbon

① Home Tab

Contains different groups

① clipboard

⑤ styles

② Font

⑥ cells

③ Alignment

⑦ Editing

④ Number

Clipboard group

Ⓐ copy - copies selected data

Ⓑ paste - pastes copied / cut data.

Ⓒ cut - moves data from one place to another

Ⓓ Format painter - copies formatting from one cell to another.

Ex:- same color / font / style

Font group

Ⓐ Font style - change text font

Ⓑ Font size - change size of text

Ⓒ Increase Font size - also: ctrl + shift + >

Ⓓ Decrease Font size - also: ctrl + shift + <

- Ⓒ Bold $\text{ctrl} + \text{B}$
- Ⓓ Italic $\text{ctrl} + \text{I}$
- Ⓔ Underline $\text{ctrl} + \text{U}$
- Ⓕ Borders - add borders to cells (used for tables)
- Ⓖ Fill color - change cell background color.
- Ⓗ Font color - change text color

Alignment Group

→ used to change text position in cell.

- Ⓐ Top Align
- Ⓑ Middle Align
- Ⓒ Bottom Align
- Ⓓ Left Align
- Ⓔ Center Align
- Ⓕ Right Align
- Ⓖ Orientation: Rotate text

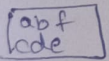
Use: long text / angle text

→ have many options like angle counter clock wise, angle clockwise, vertical text, Rotate Text up, Rotate Text down

→ can also format cell alignment (advanced alignment setting)

Ⓗ Wrap text

→ used to show long text in multiple lines inside same cell.



Ⓔ Merge & center

→ used to combine multiple cells into one cell & place text in center

Use: headings, titles, table names

Steps:-

4 options

- 1. select multiple cells
- 2. click merge & center.
- all selected cells become one cell.
- Merge & center
- Merge Across (merge row wise)
- Merge cells (merge without center)
- Unmerge cells (separate merged cells)

Number Group

→ used to change number format in cells

① There is a format bar (white box) which contains number formats formats available :-

→ General (default format of cell)

→ Number (used for normal numbers)

→ Currency (used for money values)

→ Accounting (used for financial data, proper alignment)

→ Short Date (shows date in short format)

→ Long Date (shows full date format)

→ Time (used for time values)

→ Percentage (multiplies value by 100 & adds %)

→ Fraction (shows number as fraction)

→ Scientific (shows number in power format) Ex: $1.0E+03$

→ Text (Treats value as text) & (no calculation)

→ More number formats

* Used for advanced formatting

* Custom (inside more number formats) used to create own format

Ex: ₹ #, ##

0.00

② Extra options. (Shortcuts)

→ Accounting Number format

→ Percentage style

→ Comma style Ex: 100000 $\Rightarrow$ 100,000

→ Increase Decimal Ex: 10 $\Rightarrow$ 10.00

→ Decrease Decimal Ex: 10.567 $\rightarrow$ 10.56

styles Group
cells Group } advanced later (will study further)

Editing Group

→ used for finding, editing and calculating data.

① Autosum

→ used to quickly add numbers.

- steps
1. select cells
 2. click autosum

② Fill

→ used to fill data automatically

① fill down → copies data to below cells

② fill right → copies data to right side

③ fill up → copies data upward

④ fill left → copies data to left side

⑤ across worksheets → copies data to same cell (also range) in multiple sheets

Ex: sheet 1 A1 → copied to sheet 2 A1

Steps

→ select first sheet tab

→ press ctrl

→ click another sheet tab (1 or multiple)

→ Now sheets are group

→ select data

→ click across worksheets

⑥ Series → fills number pattern

Ex: 1, 2 → fill → 3, 4, 5

Jan → Feb → Mar

⑦ Justify → used to split long text into multiple cells

→ moves text from one cell into many cells below.

→ works only for text

Steps (works only downward)

1. A1: This is a long sentence in one cell

2. select justify, Text is divided into rows

⑧ Flash fill → used to fill data automatically based on pattern.

→ Excel detects patterns & fills remaining cells

Shortcut: Ctrl + E.

Ex:

A	B
Jeevan Nallu	Jeevan
Akshaya Gunda	
Siri Nukala	
Thannu Kocheda	

select
Flash fill →

A	B
Jeevan Nallu	Jeevan
Akshaya Gunda	Akshaya
Siri Nukala	Siri
Thannu Kocheda	Thannu

→ works for text, numbers, dates

⑨ Clear -

→ used to remove data, format or links from cells

① Clear all - removes everything from cell (cell becomes empty)

② Clear formats - removes only formatting

→ removes color, font, border, style

→ data stays

③ Clear contents - removes only data

→ format stays

Ex:- Number removed but color stays.

④ Clear comments & Notes

→ data stays

⑤ Clear hyperlinks

→ Text stays

⑩ Sort & Filter

→ used to arrange & filter data

① Sorts A to Z - sorts data in ascending order

② Sorts Z to A - sorts data in descending order.

③ Custom sort - used for advanced sorting

→ can sort by column, values, color, multiple columns

④ Filter → shows only selected data

→ adds filter button on column

Use: show specific values
hide unwanted data

5) clean - Removes sorting / filter
→ data becomes normal

6) Reapply → used to apply same filter again after data changes
→ when you change data, filter does not update automatically
→ so reapply updates filter

Ex:-	Teelvani	- 50		Jiri	- 60
	Ab	- 40	$\xrightarrow{\text{filter}}$		
	siri	- 60	200	Thansri	- 60
	Tham	- 60	marks = 60		

Then I change siri marks to 70 but still it shows same

siri - 70	$\xrightarrow{\text{reapply}}$	Thansri = 60
Thansri - 60		

2) find & select

→ used to find, replace and select data

1) Find: used to search data (Ctrl + F)

2) Replace: used to change data (Ctrl + H)

3) Go to: moves to specific cell (Ctrl + G) Ex:- Go to A100

4) Go to special: select special cells Ex:- blanks, formulas, constants

5) Formulas: select all formula cells

6) Notes: select cells with notes

Use: find comments

7) Conditional formatting: select cells with conditional formatting

8) Constants: select cells with values only. Use: find number / text

9) data validation: select cells with validation rule

10) select objects: select shapes / charts

11) selection pane: shows all objects list

② Insert Tab

→ used to add elements into sheet

→ contains groups like

- | | |
|-----------------|------------|
| ① Tables | ⑥ Links |
| ② Illustrations | ⑦ Comments |
| ③ Charts | ⑧ Text |
| ④ Sparklines | ⑨ Symbols |
| ⑤ Filters | |

Tables group

• pivot table → summarize data

• Table → organize data

} covered deeply later

→ used for data structure + analysis

Illustrations group

• Pictures, shapes, icons, Smart Art

→ used for design (not important for analysis)

charts group

• column, pie, line etc..

• Pivot chart

} covered deeply later.

→ used for data visualization

sparklines group

→ Mini charts inside cells

→ used for quick trends (rarely used)

filters group

• Slicer!

• Timeline!

→ used for interactive filtering

links group

→ used to connect sheets & websites using hyperlink

Comments group

• add notes

→ used for communication

Text group

- Text box, header/footer

→ used for extra text / display

symbols group

- equations, symbols

→ used for special characters (not important)

③ Draw tab (rarely used, not asked in interviews)

→ used to draw and write on Excel sheet

what is there

- Pen - draw/write
- Highlighter - highlight context
- Eraser - remove drawing
- Ink to shape - convert drawing to shape

what we do : used for

- marking important points
- Explaining things
- annotating data

④ Page layout Tab

→ used for printing and page setup.

what is there

- Margins
- Orientation
- Size

what we do : prepare sheet for printing

⑤ Formulas Tab

→ Perform calculations on data.

function library group

what is there :

- | | | |
|-------------------|---------------|----------------------|
| • Insert function | • logical | • lookup & reference |
| • auto sum | • Text | • Math & Trig |
| • financial | • Date & time | |

- we can apply formulas like SUM, AVERAGE, IF, COUNT, VLOOKUP, XLOOKUP
- used for data calculations & logic.

Defined Names Group (not important)

- Name Manager
 - Define Name.
- Give names to ranges
 - Makes formulas easier

Formula Auditing Group

- Trace precedents
- Trace dependents
- Error checking
- Evaluate formula

What we do :- debug & check formulas

Calculation Group

- Calculate Now
- Calculate sheet
- calculation options

What we do : Control when Excel calculates formulas

Data Tab

- Contains different groups:

Get & Transform Data Group

- Used to import data from different sources
- Get Data :- from various sources like file, database, Azure etc.
- From Text / CSV :
- From web :
- From table / Range : Import data from an existing Excel table or range.
- Recent sources : shows recently used data connections sources
- Existing connections : Shows saved data connections

Queries & Connections Group

- used to manage imported data
- Queries: shows imported queries
- Connections: shows data connections
- Refresh all: Refreshes all imported data

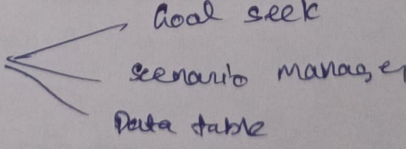
Sort & Filter Group

- used to arrange and filter data
- sort A to Z & sort Z to A
- Sort
- filter
- Advanced

Data tools Group

- used for data cleaning & preparation
- Text to Columns: splits data like Tejavani Reddy into Tejavani / Naller
- Flash fill
- Removes duplicates
- data Validation: restricts user input
- Consolidate: combines data from multiple ranges
- Relationships: creates relationship between tables

Forecast Group

- used for prediction & analysis
- what-if analysis 
 - Goal seek
 - Scenario manager
 - Data table
- forecast sheet: predict future values using existing data

Outline Group

- used to organize large data
- Group: combines rows / columns
- Ungroup: Removes grouping
- Subtotal: calculates subtotal for grouped data.